

SAI MANI KIRAN BANDI

AI/ML ENGINEER

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github.com/sbandi5

sbandi5.github.io/portfolio

OBJECTIVE

AI/ML Engineer with 2+ years of experience building and deploying Generative AI, Agentic AI, and RAG systems in production. Skilled in designing RAG pipelines, fine-tuning LLMs (LLaMA 3, Mistral, GPT-4) using LoRA/QLoRA, and building multi-agent workflows with LangChain, LlamaIndex, AutoGen, and CrewAI. Hands-on with vector databases (Pinecone, FAISS, Weaviate), LLM evaluation tools (RAGAS, TruLens), and responsible AI practices including NeMo Guardrails and PII masking. Deployed AI systems on AWS (SageMaker, Bedrock) and Azure (AI Foundry, AKS). Strong MLOps foundation with Docker, Kubernetes, MLflow, and CI/CD pipelines.

TECHNICAL SKILLS

Languages & Scripting:	Python, SQL, Bash, Scala
Generative AI & LLMs:	GPT-4, GPT-4o, LLaMA 3 (8B/70B), Mistral 7B, Claude 3, Gemini Pro, OpenAI API, Anthropic API, Hugging Face Transformers
Agentic AI & Orchestration:	LangChain, LlamaIndex, AutoGen, CrewAI, LangGraph, Semantic Kernel, Multi-Agent Systems, Tool Calling, ReAct Prompting
RAG & Vector Databases:	Retrieval-Augmented Generation (RAG), Pinecone, FAISS, Weaviate, pgvector, Azure AI Search, Chroma, Hybrid Search, BM25, Re-ranking
Fine-Tuning & Alignment:	LoRA, QLoRA, PEFT, SFT, RLHF, Prompt Engineering, Chain-of-Thought, Few-Shot Prompting, Self-Consistency
LLM Evaluation & Safety:	RAGAS, TruLens, DeepEval, BLEU/ROUGE, NeMo Guardrails, Azure Content Safety, Hallucination Detection, PII Masking, Responsible AI
ML Frameworks & Libraries:	PyTorch, TensorFlow, Scikit-learn, Keras, XGBoost, LightGBM, Pandas, NumPy, SpaCy, NLTK, OpenCV
MLOps & Deployment:	MLflow, Ray Serve, vLLM, CI/CD Pipelines, Model Monitoring, Drift Detection, A/B Testing, Automated Retraining, Model Registry
Cloud Platforms:	AWS (SageMaker, Bedrock, EC2, S3, Lambda, EKS), Azure (AI Foundry, Azure ML, AKS, Azure OpenAI, Event Hubs, Blob Storage)
Data & Big Data:	Apache Spark, Databricks, Apache Kafka, Delta Lake, ETL Pipelines, Feature Engineering, Azure Data Factory
Containers & DevOps:	Docker, Kubernetes, Helm, ArgoCD, GitHub Actions, Jenkins, Terraform, Azure DevOps
Databases & APIs:	PostgreSQL, MySQL, MongoDB, Redis, FastAPI, Flask, REST APIs, gRPC
Monitoring & Observability:	Prometheus, Grafana, OpenTelemetry, Azure Monitor, MLflow Tracking

PROFESSIONAL EXPERIENCE

AI/ML Engineer Connex AI	Sep 2025 – Present <i>Remote</i>
- Built an AI-powered credential verification microservice using Kotlin, Spring Boot, and LLM-based document classification to automatically extract, validate, and cross-reference user credentials from unstructured documents — collapsing a 2–4 week manual review process to under 1 hour for up to 500 concurrent users per batch. - Designed a RAG pipeline using LangChain and FAISS to retrieve relevant credential templates and verification rules from a structured knowledge base, improving document matching accuracy by 34% and reducing false rejection rate by 41%. - Fine-tuned a BERT-based classification model using LoRA on 80K+ credential documents for named entity recognition — achieving 93% field-level extraction accuracy across license numbers, expiry dates, issuing authority names, and credential types.	

- Debugged and resolved a Redis cache connectivity failure causing 30-second API response delays using ML-assisted log analysis — cutting average endpoint latency from 30s to 600ms and restoring real-time delivery for 500+ concurrent users.
- Designed and implemented 15+ REST API endpoints integrating the LLM verification pipeline with third-party verification providers and internal PostgreSQL databases, enabling real-time credential status updates and automated database writes on successful AI validation.
- Secured all AI service endpoints and credential data with JWT-based authentication and RBAC; wrote JUnit and Mockito tests covering core ML workflow regressions to ensure model output reliability pre-release.
- Containerized the full AI inference pipeline using Docker for consistent local, staging, and cloud Linux deployments; integrated into CI/CD via GitHub Actions with automated RAGAS evaluation gates for continuous model quality assurance.

Machine Learning Engineer
Infosys Limited

April 2022 - Jun 2023
Hyderabad, India

- Developed NLP and LLM-powered document intelligence solutions for a B2B SaaS fintech product, processing 500K+ financial and legal documents monthly across 80+ enterprise clients.
- Built Retrieval-Augmented Generation (RAG) systems using LangChain and FAISS over 400K+ indexed financial contracts and compliance documents — reducing document review turnaround by 38% and improving clause retrieval precision by 29%.
- Fine-tuned BERT and GPT-3.5 models using LoRA and SFT on 100K+ proprietary financial documents for named entity recognition, clause classification, and risk flagging — improving F1-score from 0.69 to 0.86.
- Implemented prompt engineering strategies including chain-of-thought, few-shot, and template-based prompting — improving GPT-3.5 structured data extraction accuracy by 27% without additional fine-tuning.
- Built MLOps pipelines on AWS SageMaker using MLflow for experiment tracking, model versioning, and data-drift-triggered automated retraining — reducing model degradation incidents by 55%.
- Deployed containerized ML inference services using Docker and AWS EKS behind FastAPI, scaling from 5K to 300K+ daily inference requests with zero-downtime rolling deployments.
- Developed customer churn prediction models using XGBoost and LightGBM with advanced feature engineering on 12-month behavioral data — achieving AUC of 0.88 (up from 0.72) and enabling proactive retention outreach for at-risk accounts.
- Processed and transformed 2TB+ of unstructured text data using PySpark and AWS Glue, building reusable ETL feature pipelines that reduced data preparation time by 45%.

EDUCATION

Master of Science in Computer Science, Eastern Michigan University, Ypsilanti, MI Aug 2024 – May 2026

Academic Pathway Program, Computer Science, Eastern Michigan University,
Ypsilanti, MI Aug 2023 – Apr 2024

Bachelor of Science in Electronics and Communication Engineering, Bharath Institute of Higher Education
and Research Centre, Chennai, India 2018 - 2022